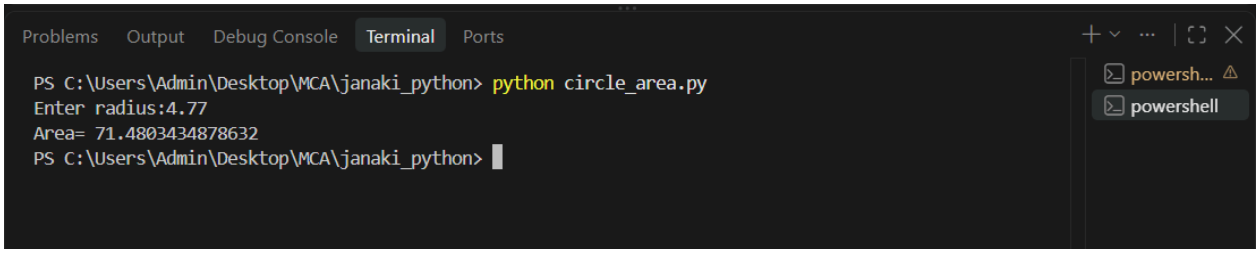
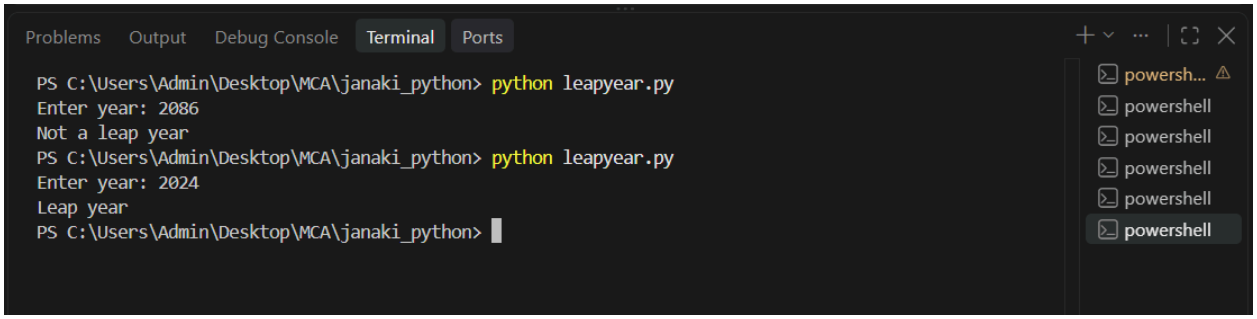
CIRCLE AREA



LEAP YEAR



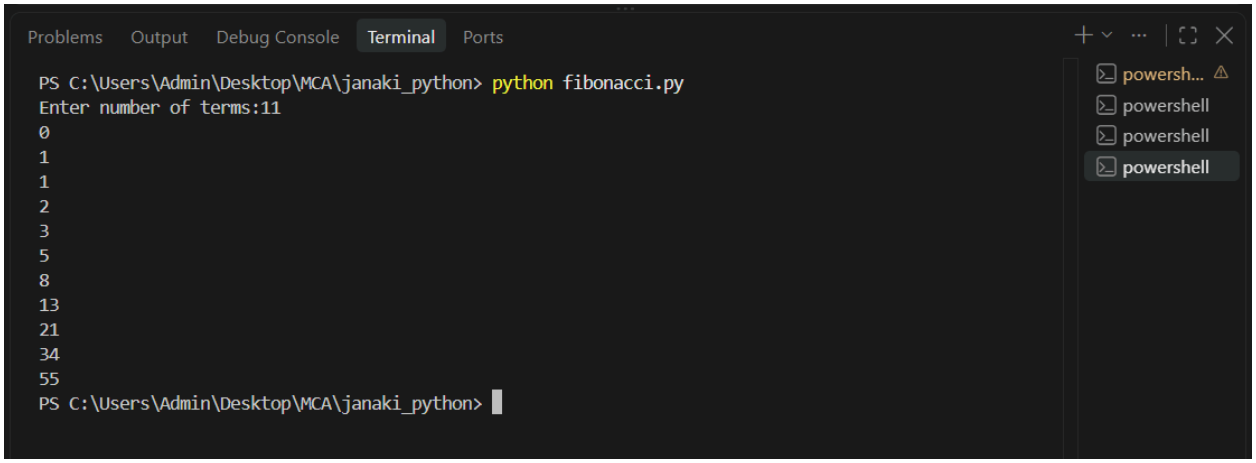
LARGEST OF THREE



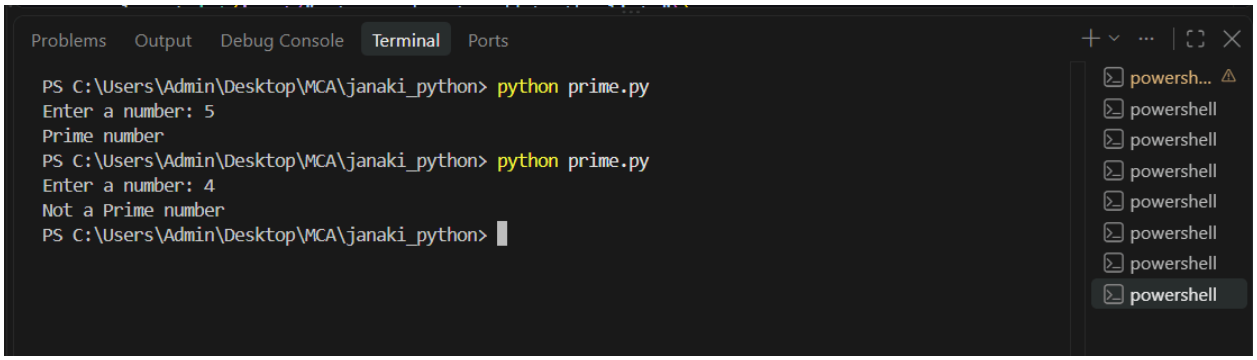
FACTORIAL



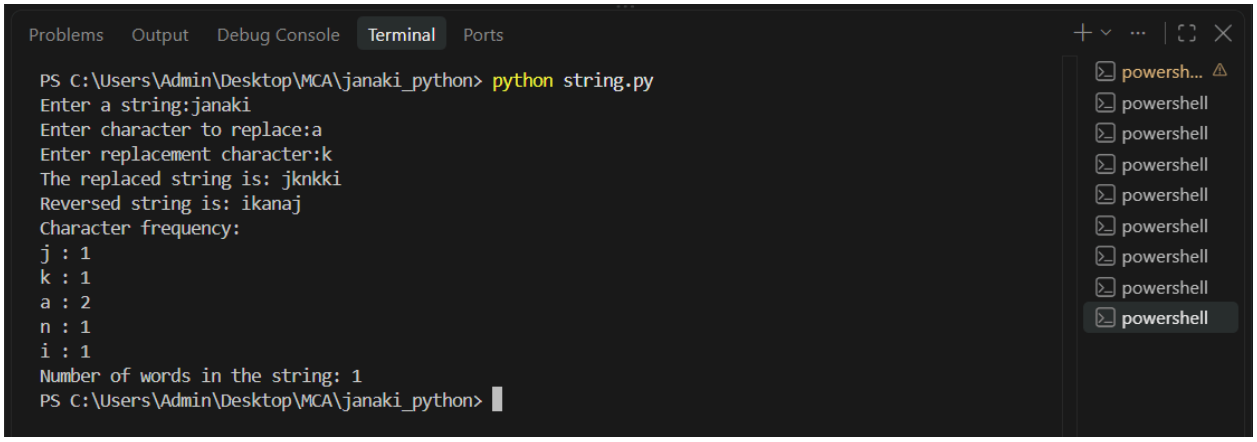
FIBONACCI



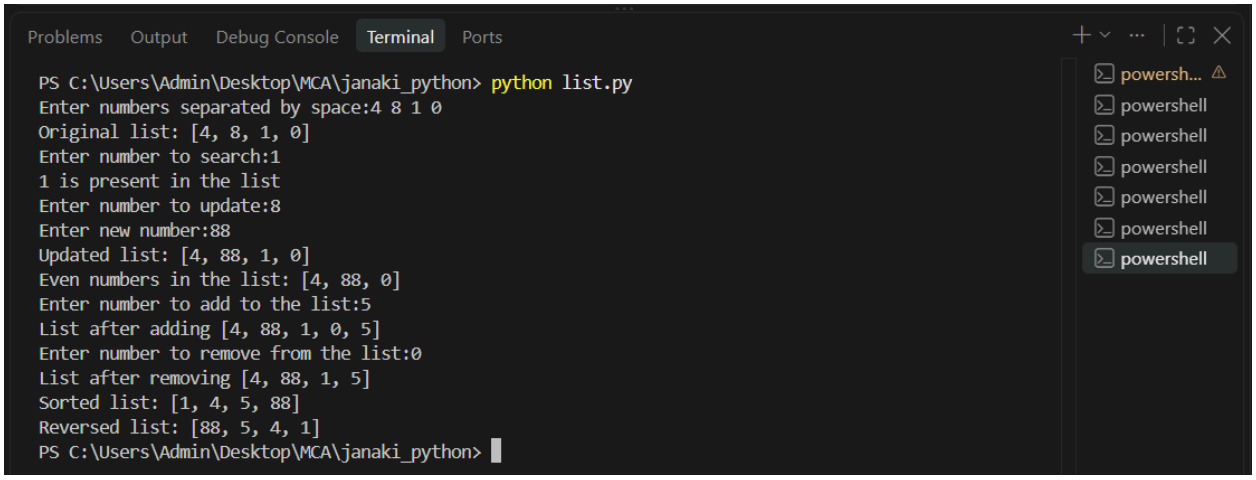
PRIME NUMBER



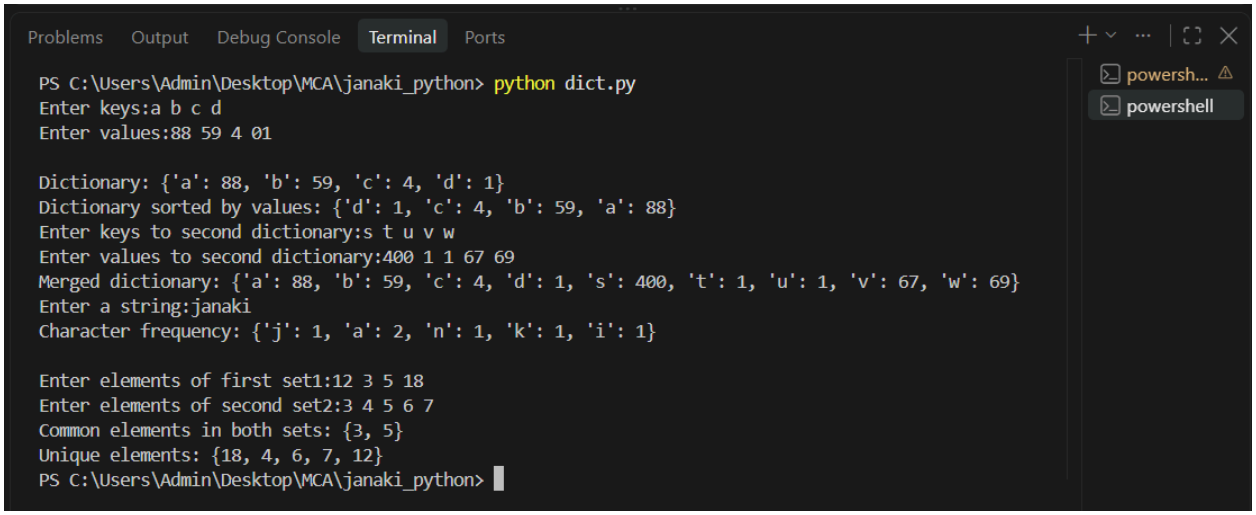
STRING



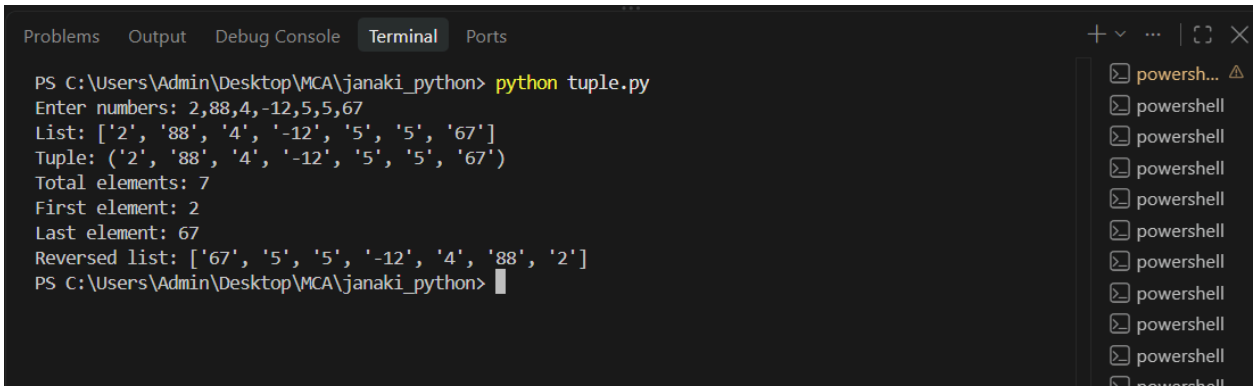
LIST



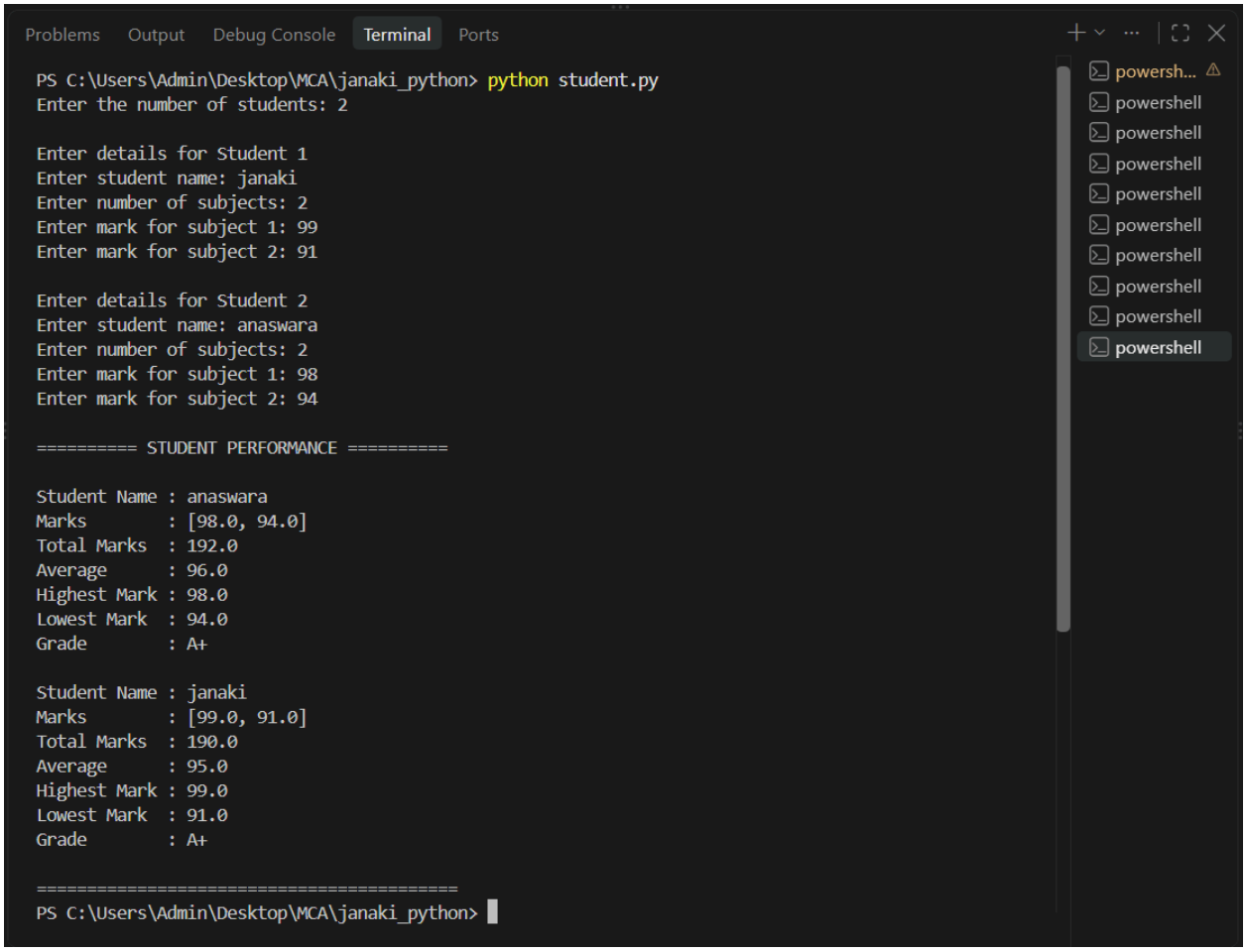
DICTIONARY



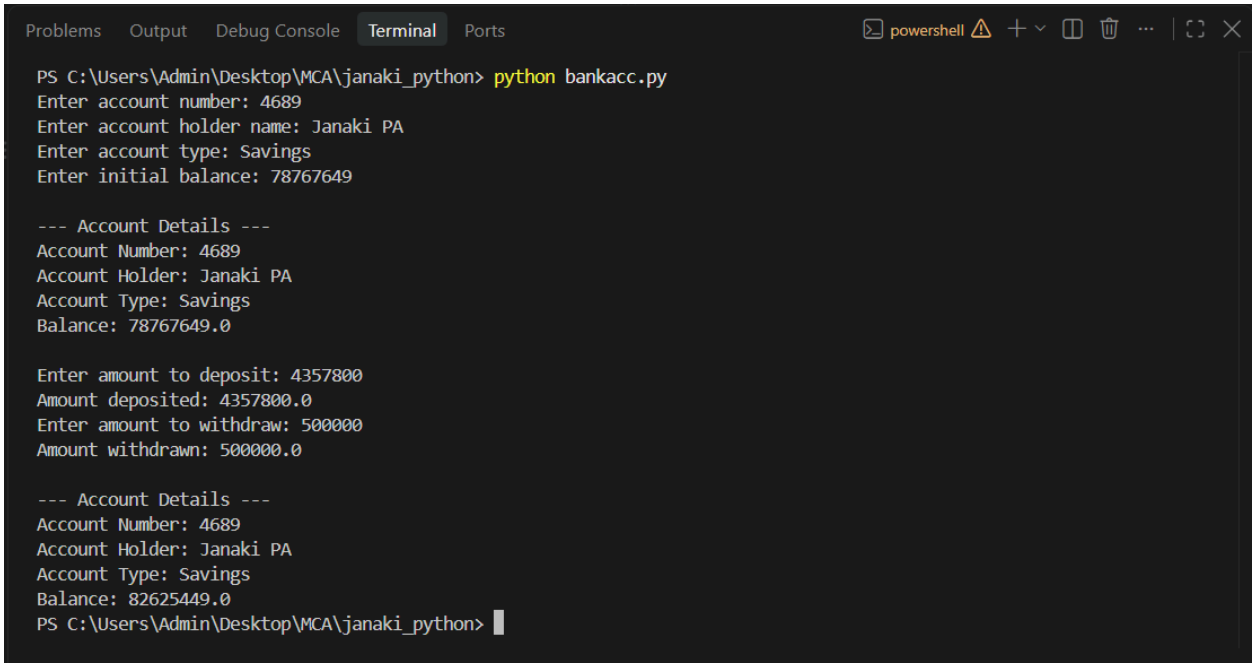
TUPLE



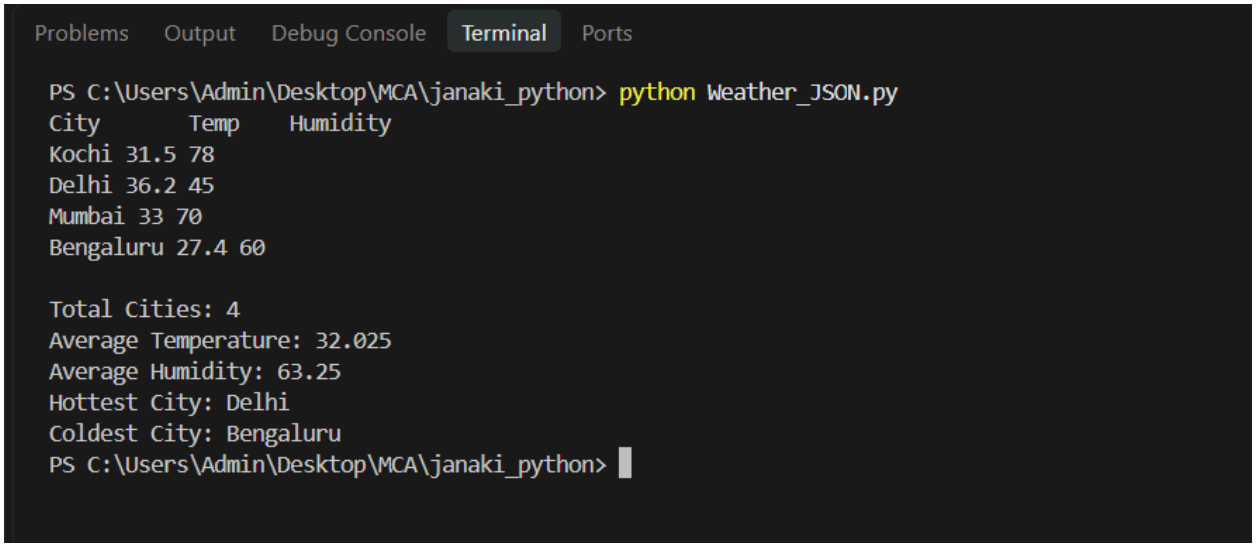
STUDENT MARKS



BANK ACCOUNT



WEATHER



ELECTRICITY BILL

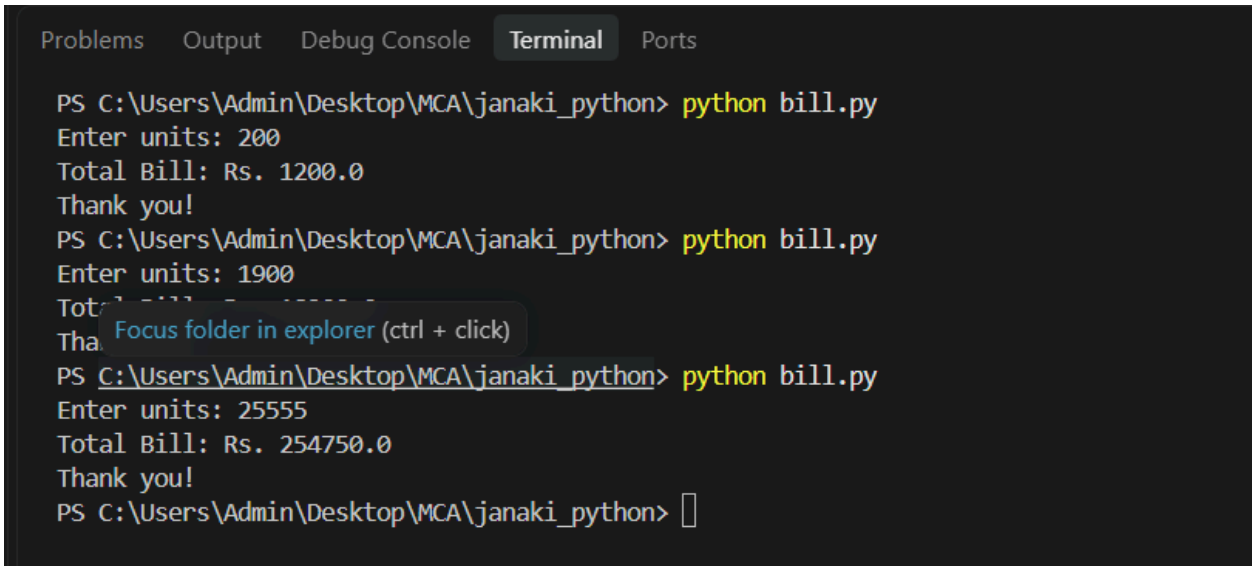


IMAGE PROCESSOR

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Admin\Desktop\MCA\janaki_python> python image.py
Original Image:
 [[ 10  50 100 150]
 [ 20  60 110 160]
 [ 30  70 120 170]
 [ 40  80 130 180]]
Reshaped:
 [[ 10  50 100 150  20  60 110 160]
 [ 30  70 120 170  40  80 130 180]]
Flattened:
 [ 10  50 100 150  20  60 110 160  30  70 120 170  40  80 130 180]
Centre Region:
 [[ 60 110]
 [ 70 120]]
First Row: [ 10  50 100 150]
Last Column: [150 160 170 180]
Bright Image:
 [[110 150 200 250]
 [120 160 210 255]
 [130 170 220 255]
 [140 180 230 255]]
Dark Image:
 [[  0   0  50 100]
 [  0  10  60 110]
 [  0  20  70 120]
 [  0  30  80 130]]
Horizontal Flip:
 [[150 100  50  10]
 [160 110  60  20]
 [170 120  70  30]
 [180 130  80  40]]
Vertical Flip:
 [[ 40  80 130 180]
 [ 30  70 120 170]
 [ 20  60 110 160]
 [ 10  50 100 150]]
PS C:\Users\Admin\Desktop\MCA\janaki_python>
```

LIBRARY

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Admin\Desktop\MCA\janaki_python> python library_Catalogue.py

1. Add Book
2. Search Book
3. Total Books
4. Books by Author
5. Exit
Enter choice: 1
Enter title: Wings of Fire
Enter author: APJ Abdul Kalam
Book added successfully

1. Add Book
2. Search Book
3. Total Books
4. Books by Author
5. Exit
Enter choice: 1
Enter title: Rathri 12 n shesham
Enter author: Akhil P Dharmarajan
Book added successfully

1. Add Book
2. Search Book
3. Total Books
4. Books by Author
5. Exit
Enter choice: 2
Enter title or author: apj abdul kalam
2 Wings of Fire by APJ Abdul Kalam

1. Add Book
2. Search Book
3. Total Books
4. Books by Author
```

Problems Output Debug Console Terminal Ports

```
1. Add Book
2. Search Book
3. Total Books
4. Books by Author
5. Exit
Enter choice: 4
Enter author name: apj
Wings of Fire by APJ Abdul Kalam

1. Add Book
2. Search Book
3. Total Books
4. Books by Author
5. Exit
Enter choice: 3
Total books: 3

1. Add Book
2. Search Book
3. Total Books
4. Books by Author
5. Exit
Enter choice: 4
Enter author name: akhil
Rathri 12 n shesham by Akhil P Dharmarajan

1. Add Book
2. Search Book
3. Total Books
4. Books by Author
5. Exit
Enter choice: 5
PS C:\Users\Admin\Desktop\MCA\janaki_python>
```

INPUT VALIDATION

